Aryan Miriyala

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Professional Summary

Computer science graduate student and software engineer with experience across full-stack product engineering, applied AI, cloud data engineering, secure workflow modernization, cybersecurity education, manufacturing/process-improvement technology, and systems research. Built customer/admin portals, LLM/RAG workflows, AI-assisted products, AWS data pipelines, database automation, secure internal tools, healthcare workflow improvements, and C++/OpenMP runtime experiments using Python, TypeScript/JavaScript, SQL, React, Next.js, Firebase, Angular, Node.js, PostgreSQL, MongoDB, MySQL, Docker, AWS, PySpark, LangChain, OpenAI API, Gemini API, Mistral AI, and ChromaDB.

Education

Bowling Green State University — Bowling Green, OH
Master of Science in Computer Science, GPA: 4.00/4.00 — *Aug. 2024 – Aug. 2026*

Bowling Green State University — Bowling Green, OH
Bachelor of Science in Computer Science, GPA: 4.00/4.00 — *Jan. 2021 – Apr. 2024*

Experience

Software Engineering Intern — Jul. 2026 – Present
Actual Reality Technologies — *Toledo, OH*

- Improving a private Next.js/TypeScript customer portal hosted at my.actualreality.ai, spanning customer/admin workflows, Firebase-backed auth/data paths, Plane-backed project data, dashboard status logic, health checks, and Vitest/Testing Library coverage.
- Consolidated separate Capture, Define, and Whiteboard tools into a feature-gated tabbed Workshop route with legacy redirects, reusable tab UI, ARIA/tab-history guardrails, route tests, component tests, and documentation updates.
- Hardened Plane-backed project reliability by paginating work-item state counts, forcing cache-bypassing refreshes through route/gateway layers, preventing stale overlapping reads, and deriving project states from issue counts instead of rounded progress.
- Improved production authentication and session behavior by deriving cookie domains from request hosts, stabilizing password and magic-link login transitions, and working through follow-up account-lifecycle risks around magic-link account creation, role-change side effects, invite domains, and Auth/user-document drift.
- Added and extended Vitest, Testing Library, route, component, architecture, and health-check coverage for Plane integration, dashboard semantics, auth/session behavior, membership remediation, revoked write access, Firestore read-error visibility, and customer workflow validation.

Artificial Intelligence & Computer Science Intern — May 2026 – Jul. 2026
SmartSolve Industries — *Bowling Green, OH*

- Worked under a phased AI integration roadmap that progressed from environment onboarding and quick-win deployments toward proprietary internal tool development, strategic AI planning, documentation, and stakeholder-facing technical handoff.
- Architected a full-stack onboarding tracker with Next.js, TypeScript, PostgreSQL, Drizzle ORM, SSO, and auth middleware to centralize HR/new-hire workflows and protect sensitive employee data.
- Supported AI adoption planning by auditing AI tool usage opportunities, identifying workflow gaps and redundancies, and translating department-level business needs into candidate AI capstone projects with expected outcomes and success criteria.
- Scoped an AI-enabled QMS dependency register for quality-management documents, modeling document relationships and converting ambiguous operational needs into a reviewable implementation plan.
- Built a cross-platform Docker devcontainer for WSL2 and Colima, improving reproducible setup and isolating AI-assisted development workflows around proprietary source code.
- Evaluated LLM/API and cloud-service integration paths for internal workflow automation, including lightweight proprietary tools, API integrations, webhook-based automations, internal knowledge-base documentation, and applied business-problem training material.
- Applied Codex and Claude Code for planning, implementation, debugging, code review, and self-review while keeping human validation central around proprietary source code and company data.

Graduate Research Assistant I — Aug. 2024 – May 2026
Bowling Green State University — *Bowling Green, OH*

- Developed cybersecurity education materials, lab guides, workshops, and presentations across ethical hacking, OSINT, web security, email security, cryptography, secure key management, and cyber workforce topics.

- Built hands-on instructional content using Kali Linux, Nmap/NSE, Metasploit, OpenVAS/Greenbone, Wireshark/tcpdump, Burp Suite, OWASP ZAP, Shodan, Netcat/Ncat, Snort, John the Ripper, Hydra, DVWA, Metasploitable2, Vulnserver, and PentesterLab.
- Supported NICE Framework and CAE-CD curriculum analysis by mapping cybersecurity roles, knowledge areas, tools, and instructional requirements into student- and instructor-facing materials.
- Created email security and forensics material covering phishing analysis, email headers, SPF/DKIM/DMARC, PGP/GnuPG, packet captures, and secure email communication.
- Built cryptography and secure key-management content involving symmetric/asymmetric encryption, hashing, PKI, key storage, key recovery, Azure Key Vault lab material, and Shamir's Secret Sharing.
- Led 10 undergraduate students in organizing a middle-school CyberCamp using CyberGuardian modules, cryptography activities, phishing demos, password-security exercises, and cyber escape-room material.

Data Engineering Intern May 2025 – Aug. 2025
American Association of Insurance Services *Lisle, IL*

- Automated a fully manual SQL billing workflow by building a production PySpark job and AWS Glue Workflow, processing 20+ TB of golden-table insurance data to calculate charges for 700+ member companies.
- Profiled 160+ MySQL, Oracle, and Impala tables using Python, Pandas, SQL, and JDBC to map source-system similarities and define a unified taxonomy across 25 MDM domains.
- Replaced 160+ legacy Pentaho jobs with AWS Glue ETL workflows and partitioned S3 pipelines for Semarchy MDM, improving maintainability and standardizing cloud data movement.
- Built reverse-ETL and data-validation workflows with custom AWS IAM roles to support controlled self-service data access, reduce manual entry, and maintain 24-hour data latency.
- Worked across Python, PySpark, AWS Glue, Glue Workflows, S3, IAM, SQL, MySQL, Impala, Oracle, PostgreSQL, JDBC, Semarchy MDM/xDM, golden tables, billing workflows, and insurance data platforms.

Software Engineering Intern May 2023 – May 2024
American Association of Insurance Services *Lisle, IL*

- Engineered openIDL/openIDS-aligned data-modeling workflows by parsing insurance taxonomies with Python/JSON logic and generating 1,000+ production SQL tables across MySQL, Oracle, PostgreSQL, and 10+ insurance lines.
- Built Python data-processing workflows with Pandas, multiprocessing, fuzzy matching, and hashing to clean large CSV datasets and support insurance data-standardization pipelines.
- Contributed to a full-stack React/Node.js modernization by implementing RBAC and JWT authentication patterns to replace broad employee-wide data access with role-specific controls for about 110 users.
- Automated database validation with AWS Lambda triggers that tokenized PII using hashlib/boto3 and ingested structured JSON into S3 for downstream standardization workflows.
- Worked across Python, JSON, Pandas, SQL, MySQL, Oracle, PostgreSQL, React, Node.js, JWT, RBAC, AWS Lambda, AWS S3, boto3, hashlib, large CSV processing, and insurance taxonomies.

Software Engineering Intern May 2022 – Aug. 2022
Alliance for Paired Kidney Donation *Toledo, OH*

- Improved an AWS-hosted Lucee/CFML healthcare workflow platform for kidney-paired donation operations by strengthening form security, auditability, data-entry reliability, and internal clinical workflow usability.
- Implemented global anti-CSRF form protection, IP-aware audit logging, protected access-denied routing, and clearer 404 behavior across internal workflows.
- Modernized legacy ColdFusion/Lucee forms with HTML5 date/time pickers, themed CSS, JavaScript fallbacks, cfform script updates, and graceful degradation for manual input.
- Improved transplant logistics and data reliability by adding center-level blood-tube defaults, logistics-sheet population, Matchgrid upload ordering, MFI file-location fixes, and duplicate-antigen MFI crash handling.
- Refactored MatchRun, registration, offer, and combination interfaces into reusable modal/tab components with icons and tooltips to improve usability and maintainability.

Projects

Travel Health Advisor - BGSU Hackathon 2025 2025

- Built a full-stack health AI prototype with React, Vite, Express, MongoDB/Mongoose, and Mistral AI for travel-health guidance.
- Implemented an interactive country-selection map with react-simple-maps, retrieving disease records from an Express/MongoDB API and surfacing risk, vaccine availability, healthcare access, recovery-rate data, and travel recommendations.
- Built a health-profile form and context-aware Mistral AI chatbot that used saved user health details to answer travel-health questions.

- Generated downloadable vaccination checklist PDFs with @react-pdf/renderer based on country-specific vaccine-required disease data.

Fix-It-Flow - RocketHacks 2026 2026

- Built a Next.js 14/TypeScript PWA that uses camera input, voice commands, Gemini Vision, Featherless.AI/Llama reasoning, ElevenLabs TTS, and DynamoDB to guide appliance repair and recycling decisions.
- Implemented a voice-first inspection workflow combining live-frame analysis, spoken user context, conversation history, structured findings, confidence scores, requested camera views, and safety warnings.
- Designed hands-free repair guidance with AI-generated steps, ElevenLabs audio playback, contextual Q&A, and voice commands for next, repeat, help, and end session.
- Built typed Next.js API routes and AWS-backed session persistence for inspection turns, repair sessions, device identification, text-to-speech, chat, uploads, findings, frames, and repair steps.

Diff-Grounded Pull Request Description Generation with Structured Evidence 2026

- Built an ICSME-published LLM research pipeline that generated evidence-grounded GitHub pull request descriptions from commits, file diffs, linked issues, and repository metadata.
- Developed structured PR-context construction, weak-commit-message improvement, file-level summarization, and grounding-constrained generation workflows using Python, OpenAI API, and GitHub API.
- Built an automated LLM evaluation workflow comparing generated PR descriptions against raw code-change evidence, outperforming AIDev and PRSummarizer baselines in correctness, coverage, and clarity.
- Used Python, OpenAI API, LLM inference, prompt engineering, GitHub API, retrieval-augmented context construction, knowledge graphs, software repository mining, NLP, pandas, and scikit-learn.

DreamScape: AI-Powered Sleep Learning Companion

- Built a cross-platform React Native/Expo mobile app for sleep-based microlearning with AI-generated study cues, quizzes, text-to-speech audio, and local-first session tracking.
- Integrated Gemini and ElevenLabs APIs to summarize PDF/TXT documents, generate quizzes, synthesize learning cues, cache audio, and schedule simulated sleep-mode playback.
- Designed typed Zustand/AsyncStorage stores for topics, flashcards, cues, sleep sessions, quiz results, and user settings in a local-first mobile architecture.
- Used Expo SDK 54, Expo Router, TypeScript, Gemini API, ElevenLabs API, Expo AV, Expo FileSystem, Expo Document Picker, Zod, and React Navigation.

FalconGraph Search: AI-Powered Campus Knowledge Search

- Built an AI-powered BGSU campus search prototype combining lightweight RAG, live web search, crawled webpages, PDFs, and LLM inference to generate source-grounded answers.
- Implemented document ingestion, chunking, vector retrieval, semantic search, and source-aware summarization across stored and crawled university resources.
- Developed graph-search functionality and an interactive graph interface using page/resource relationships to trace generated answers back to original sources.
- Used Python, Flask, OpenAI API, embeddings, vector retrieval, semantic search, web crawling, PDF parsing, React, Next.js, Three.js, Tailwind CSS, C++, graph algorithms, and pandas.

HealthTrend: Healthcare Big Data Streaming Pipeline

- Built a Dockerized healthcare data pipeline using Apache NiFi, Kafka, Hadoop HDFS, and Spark to ingest structured patient records and stream IoT health events.
- Routed CSV patient data into HDFS and JSON health events into Kafka, then processed both sources with Spark/PySpark for validation and downstream analytics.
- Simulated a healthcare analytics architecture for patient records, heart rate, oxygen level, and temperature data inside an isolated Docker Compose environment.
- Used Apache NiFi, Apache Kafka, Apache Zookeeper, Hadoop HDFS, Spark, PySpark, Docker Compose, Python, JSON, CSV, stream ingestion, batch storage, and distributed processing.

MarioRL: Super Mario Reinforcement Learning Agent Comparison

- Built a reinforcement learning project comparing PPO and DQN agents on Super Mario Bros using Python, PyTorch, Stable-Baselines3, Gym, and TensorBoard.
- Implemented environment preprocessing with frame skipping, resizing, grayscale conversion, and frame stacking.
- Evaluated agents through checkpoints, training/testing scripts, generated clips, TensorBoard logs, and performance graphs comparing reward, episode length, and training speed.
- Used gym-super-mario-bros, NES-Py, custom agent/network code, Jupyter Notebook, experiment artifacts, and reinforcement-learning evaluation workflows.

RAG-Based Analysis of App Update Frequency and User Reviews

- Built a RAG research pipeline using Python, Sentence Transformers, ChromaDB, embeddings, vector search, and review mining to study how mobile app update frequency relates to Google Play review feedback.
- Applied semantic retrieval, sentiment analysis, JSON preprocessing, and review-mining techniques to compare maintenance patterns, recurring user concerns, and app version histories.
- Used Python, RAG, Sentence Transformers, ChromaDB, vector search, semantic retrieval, NLP, scikit-learn, JSON processing, data preprocessing, information retrieval, research evaluation, and LaTeX.

RocketGrader: Intelligent Assignment Feedback Platform

- Built a full-stack AI assignment feedback platform with Angular, TypeScript, Express, MongoDB, Auth0, AWS S3, LangChain, and Mistral AI.
- Designed REST APIs and MongoDB/Mongoose models for users, classes, assignments, submissions, files, and chat while supporting role-based teacher/student dashboards.
- Implemented submission upload, parsing, and AI grading across PDF, DOCX, HTML, and text files, returning structured scores, feedback, strengths, improvements, and rubric breakdowns.
- Used Angular 19, TypeScript, Auth0, Node.js, Express, MongoDB, Mongoose, REST APIs, JWT, bcrypt, AWS Amplify, Multer, Mammoth, JSDOM, Tailwind CSS, and DaisyUI.

Self-Adaptive Parallelism via OpenMP Runtime Coordination

- Built a C++/OpenMP adaptive runtime research system that used per-epoch telemetry and a UCB controller to tune scheduling policy, chunk size, and thread count.
- Benchmarked adaptive scheduling against serial and fixed-policy baselines across Mandelbrot, heat diffusion, and reduction workloads.
- Generated paper-ready performance summaries and plots for runtime, speedup, efficiency, Karp-Flatt behavior, and scheduler decisions.
- Used C++, OpenMP, Python, PyQt, Docker, Make, Linux, TCP servers, runtime instrumentation, UCB bandit optimization, adaptive scheduling, CSV telemetry pipelines, performance benchmarking, and data visualization.

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, C++, C, C#, PHP, Lucee/CFML, ColdFusion
Frontend: React, Next.js, Next.js 15, Next.js App Router, Angular, Vite, PWA, React Native, Expo, Expo Router, React Navigation, Tailwind CSS, DaisyUI, Bootstrap, HTML5, CSS, D3, TopoJSON, react-simple-maps, Framer Motion, Lucide React, React Icons, React Toastify, React Markdown, responsive UI, browser camera APIs, Web Speech API, modal/tab UI patterns
Backend, APIs & Auth: Node.js, Express.js, Flask, REST APIs, Axios, CORS, dotenv, auth middleware, JWT, RBAC, SSO, Auth0, Firebase Auth, Firebase Admin, Drizzle ORM, Mongoose, bcrypt, @react-pdf/renderer, Lucee/CFML backend workflows
Cloud, Data Engineering & DevOps: AWS Glue, AWS Glue Workflows, AWS S3, AWS Lambda, AWS IAM, AWS DynamoDB, AWS SDK for JavaScript, AWS Rekognition client code, PySpark, Spark, Apache NiFi, Apache Kafka, Apache Zookeeper, Hadoop HDFS, MapReduce, Impala, ETL, reverse ETL, data validation, data profiling, MDM, Semarchy MDM/xDM, partitioned S3 pipelines, golden tables, JDBC, Docker, Docker Compose, devcontainers, WSL2, Colima, Git, Linux, Make, Vercel, AWS Amplify
Databases: PostgreSQL, MySQL, MongoDB, MongoDB Atlas, Oracle, ChromaDB, Firebase/Firestore
AI, ML & Retrieval: Codex, Claude Code, OpenAI API, Gemini API, ElevenLabs API, Featherless.AI, Llama 3.1, Qwen2.5-VL, Mistral AI, LangChain, RAG, prompt engineering, LLM inference, multimodal reasoning, automated LLM evaluation, HuggingFace, Sentence Transformers, embeddings, vector search, semantic retrieval, cosine similarity search, PyTorch, TensorFlow, Stable-Baselines3, reinforcement learning, PPO, DQN, Gym, TensorBoard, scikit-learn, NLP, review mining
Security, Testing & Compliance-Aware Engineering: SSO, JWT, RBAC, auth middleware, anti-CSRF tokens, audit logging, IP-aware logging, access-denied workflows, protected 404 workflows, data access controls, PII tokenization, hashlib, boto3, healthcare/compliance-aware workflows, Vitest, Testing Library, React Testing Library, ESLint, route tests, component tests, architecture guardrail tests, Firebase rules testing, Plane API/integration
Cybersecurity Education Tooling: NICE Framework, CAE-CD curriculum mapping, CyberGuardian, Kali Linux, OSINT, Nmap/NSE, Metasploit, OpenVAS/Greenbone, Wireshark/tcpdump, Burp Suite, OWASP ZAP, OWASP Top 10, Shodan, theHarvester, Netcat/Ncat, Masscan, Snort, John the Ripper, Hydra, DVWA, Metasploitable2, Vulnserver, PentesterLab, SPF/DKIM/DMARC, PGP/GnuPG, phishing analysis, cryptography, PKI, secure key management